

NICK PARVATH

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PROFESSIONAL SUMMARY

Research-driven Machine Learning and Data Science professional specializing in **large-scale search, information retrieval, ranking, relevance, semantic understanding, and search evaluation**. Led applied search initiatives spanning **retrieval quality, ranking optimization, query-item relevance, cold-start discovery, semantic signals, human evaluation, model diagnostics, and experimentation** across production-scale international search systems. Experienced translating ambiguous search problems into measurable retrieval objectives, evaluation datasets, relevance metrics, failure analyses, model improvements, and scalable production workflows.

Strong background across **information retrieval, machine learning, statistical modeling, ranking systems, experimentation, and applied research**, with depth in offline and online search evaluation, human-labeled relevance assessment, threshold optimization, error analysis, model comparison, and A/B testing. Experienced leading cross-functional work with ML engineering, product, analytics, and platform teams to define **retrieval objectives, relevance signals, ranking strategies, evaluation methodology, quality guardrails, and research directions**. Research interests include **neural information retrieval, semantic retrieval, embedding-based search, reinforcement learning, agentic search, LLM-based retrieval systems, cold-start ranking, and search evaluation**.

EDUCATION

Northeastern University

Boston, MA

Master of Science in Engineering Management; Major: Mathematical Modeling, LLM Modeling, Product Management

RV College of Engineering

Bangalore, India

Bachelor of Technology in Electronics and Communication Engineering

SKILLS

Search & Retrieval	Information Retrieval, Ranking, Semantic Search, Relevance Modeling, Personalization, Recommendation Systems, Cold-Start Retrieval, Query-Item Relevance
Evaluation & Experimentation	Human Evaluation, Offline/Online Evaluation, Failure Analysis, A/B Testing, Precision, Recall, F1, nDCG, CTR, Threshold Analysis, Model Comparison
ML & Research	Machine Learning, Statistical Modeling, Reinforcement Learning, Generative AI, LLMs, Agentic AI, Feature Engineering, Model Diagnostics, Applied Research
Programming & Systems	Python, SQL, PySpark, Pandas, NumPy, Scikit-learn, XGBoost, Spark ML, TensorFlow, PyTorch, Java, C++, Databricks, Docker, Kubernetes
Technical Leadership	Research Problem Framing, Technical Strategy, Cross-Functional Leadership, Experiment Design, Stakeholder Alignment, Launch Planning, Technical Communication

PROFESSIONAL EXPERIENCE

Walmart Global Tech

Bay Area, CA

Data Scientist – AdTech, Search and Personalization

- Led **search, retrieval, and ranking initiatives** across international marketplaces, driving applied ML work across retrieval quality, semantic relevance, query-item matching, cold-start discovery, personalization, variant selection, and ranking optimization for Canada, Mexico, and emerging markets.
- Architect end-to-end **retrieval and ranking workflows** connecting customer intent, product representations, behavioral signals, semantic features, relevance scores, scalable data pipelines, model evaluation, experimentation, and production-readiness decisions.
- Develop **relevance and retrieval frameworks** using customer interaction data, product catalog attributes, semantic signals, pricing features, quality indicators, marketplace context, and behavioral evidence to improve which products are retrieved, retained, and prioritized within search experiences.
- Design and own **search evaluation frameworks** combining human-labeled relevance judgments, offline model scores, behavioral signals, and online experiment outcomes to evaluate retrieval and ranking quality through precision, recall, F1, nDCG, CTR, conversion, and customer-impact metrics.
- Build **human-evaluation and benchmark workflows** to compare retrieval and ranking variants, quantify relevance quality, identify regressions, analyze disagreement patterns, optimize thresholds, and uncover query- and item-level failure modes hidden by aggregate metrics.
- Drive deep **search failure analysis and model diagnostics** across query classes, product attributes, markets, semantic signals, relevance bands, ranking positions, and edge cases, translating findings into improvements to features, thresholds, ranking logic, and fallback behavior.

- Shape **cold-start retrieval and ranking strategy** by combining semantic relevance, controlled exploration, calibration, lifecycle evidence, behavioral signals, and quality guardrails to improve discovery for high-quality products with limited historical interaction data.
- Own **search experimentation strategy** by translating ambiguous retrieval and ranking questions into testable hypotheses, evaluation datasets, offline success metrics, A/B testing frameworks, statistical readouts, risk guardrails, and launch recommendations.
- Partner closely with **ML Engineering, Search Engineering, Product, Analytics, Data Platform, and international market teams** to define relevance signals, ranking objectives, evaluation methodology, production workflows, system behavior, launch criteria, and market expansion strategies.
- Led applied **search and ranking research** alongside production work, translating retrieval, relevance, personalization, and cold-start challenges into research papers, patent concepts, experimental frameworks, and longer-term ML directions.

Volvo Group

Boston, MA

AI/ML and IoT Engineer

- Led AI/ML work for the **Flowlane initiative**, focused on automating end-station workflows through computer vision, IoT-driven monitoring, real-time process intelligence, and industrial automation.
- Developed **computer vision and ML workflows** using Python, OpenCV, and TensorFlow to support automated inspection, workflow monitoring, and process understanding, achieving prototype-level detection accuracy in the **85-90% range** during controlled testing.
- Built **mathematical modeling and evaluation frameworks** to analyze takt time, dwell time, throughput, bottlenecks, process efficiency, model feasibility, scalability, and business value across manufacturing automation use cases.
- Connected **sensor data, workflow data, and operational signals** into scalable analytics frameworks, reducing manual process tracking effort by an estimated **30-40%** and improving visibility into bottlenecks, system behavior, and throughput patterns.
- Designed monitoring and evaluation workflows to help engineering and operations teams identify performance patterns, analyze system constraints, detect operational issues, and prioritize model and process improvements.
- Created analytics and visualization workflows using **SQL, Power BI, and ThingWorx** to monitor system performance, process bottlenecks, operational health, and automation readiness, reducing recurring manual reporting effort by approximately **25%**.
- Collaborated with engineering, operations, and product stakeholders to define technical requirements, validate ML outcomes, investigate system behavior, and translate AI/IoT findings into scalable workflow recommendations.

Cisco

Bangalore, India

Automation Engineer

- Supported enterprise automation initiatives involving **software engineering**, data processing, operational analytics, monitoring workflows, and large-scale technical systems.
- Built analytical workflows, automation scripts, and reporting solutions to improve **system visibility**, debugging efficiency, workflow reliability, and engineering decision support.
- Collaborated with engineering teams to translate business and technical requirements into scalable implementation workflows and production-ready solutions.
- Applied **Python, SQL, Java, and cloud tools** to support data-driven problem-solving, automation, and system analysis across large-scale engineering environments.

RECENT PATENTS AND RESEARCH PAPERS

MESA-RLSP: Multi-Modal Evidence Sequencing Allocation with Reinforcement Feedback and Surface Priors for Cold-Start Personalization

Research Paper

- Led development of a research framework for **cold-start retrieval, ranking, and personalization** using multi-modal evidence sequencing, reinforcement feedback, surface priors, semantic evidence, and adaptive exposure strategies.
- Investigated how search and ranking systems can balance **retrieval relevance, exploration, evidence quality, semantic alignment, customer intent, and downstream performance** when historical interaction signals are sparse or unavailable.

Cart Checkout Optimization via Price Gap-Aware Product Recommendation & Dynamic Threshold Modeling

Patent Pending

- Designed a patent-pending **context-aware retrieval and recommendation framework** using price-gap awareness, dynamic threshold modeling, cart context, behavioral evidence, and personalized relevance signals.
- Designed the decision framework to determine **when, why, and which candidate products** should be retrieved and surfaced, balancing relevance, customer intent, price sensitivity, cart value, and conversion likelihood.